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IONE: Identifiability-Oriented Navigation of Evidence for Auditing Aggregate Clinical Trial Data

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Abstract

Even a well-conducted randomized trial or meta-analysis publishes, at best, arm-level margins and a few subgroup tables; whether those aggregates can justify a treatment decision for a particular patient subgroup is a question of identification, not estimation. We present `ione` (IONE: Identifiability-Oriented Navigation of Evidence), an open-source Python package that audits published aggregate randomized-trial data without requiring individual patient data. For a binary stratum, binary outcome, and two-arm design, the set of conditional risk distributions compatible with the published margins is a polytope, and `ione` computes sharp bounds on the stratum-specific risk difference by linear programming (HiGHS), returning one of five explicitly disjoint verdicts — `ROBUST_TREAT`, `ROBUST_NO_TREAT`, `REVERSIBLE`, `INFEASIBLE`, or `UNSUPPORTED` — so that contradictory inputs, missing specification, and genuine nonidentification are never conflated. When a decision is not robust, the package orders additional *reportable* aggregate summaries by burden (re-tabulation before new measurement before new enrollment) without promising resolution. Applications to published data from RECOVERY (dexamethasone in hospitalized COVID-19 patients), ISIS-2 (aspirin and streptokinase in acute myocardial infarction), and the WHO REACT corticosteroid meta-analysis show that published subgroup margins are usually decision-reversible, and a retrospective leave-one-subgroup-out check on RECOVERY reproduces a realistic direction reversal that unexamined pooling would have produced. The tool recommends no treatment; it reports what the published counts identify and what reportable information would narrow the bounds.

Keywords: partial identification, aggregate data, clinical trials, decision analysis, evidence audit, Python.

1. Introduction

Clinicians and evidence-synthesis researchers routinely face a situation in which a randomized controlled trial (RCT) or meta-analysis is statistically sound, yet the published report does not settle the question that matters in front of them: whether a treatment helps the specific subgroup their patient belongs to. Trials typically report arm-level event counts for the whole sample and, occasionally, crude subgroup tables. Whether those aggregates are even consistent with a unique subgroup-level treatment effect — and, if not, how wide the range

of compatible effects is — is a question of identification, not of estimation precision.

That published margins do not uniquely determine conditional quantities is a classical result. Manski's framework of partial identification (Manski 2003) formalizes bounds on quantities that remain set-valued given the available information; Cross and Manski (2002) showed that aggregate ("short") regressions identify conditional means only up to intervals; King (1997) obtained deterministic bounds on individual-level cell probabilities from aggregate two-by-two margins in the ecological inference literature; and Horowitz and Manski (2000) derived bounds on treatment effects in randomized experiments with missing covariate and outcome data. Linear programming formulations for sharp treatment-effect bounds under structural assumptions go back to Balke and Pearl (1997). More recently, Moon (2026) independently developed sharp bounds for linear combinations of conditional mean outcomes when only group-level marginal distributions are observed — mathematically the same family of marginal-consistency programs that we use, motivated by economic outcomes. None of this work, however, addresses the applied question a clinical researcher faces at the level of a published report: given *this* trial's printed tables, is a decision for *this* subgroup already determined, could it still reverse, and which additional *reportable* summary — one the original investigators could plausibly supply — would narrow the answer?

Existing software does not close this gap. Meta-analytic packages (Viechtbauer 2010) synthesize published effects but assume the estimates are already identified. Sensitivity-analysis tools such as the E-value (VanderWeele and Ding 2017) quantify robustness to unmeasured confounding for a point-identified estimate — a different question from nonidentification. Bound software exists in econometrics — the `tebounds` command (McCarthy, Millimet, and Roy 2015) and the `bpbounds` implementation of the Balke–Pearl bounds (Palmer, Ramsahai, Holmes, Steer, Sheehan, Tobias, Lawlor, and Smith 2011) — but these require microdata or instrumental-variable designs and do not operate on published arm-level margins. Transportability theory (Pearl and Bareinboim 2014) identifies conditions under which a causal effect may be carried to a new population, but presumes assumptions that a published report rarely documents.

We present `ione` (IONE: Identifiability-Oriented Navigation of Evidence), an open-source Python package that turns this identification question into a reproducible audit. Its contributions are deliberately those of an applied tool, not of new theory: (i) a complete audit workflow on published aggregate counts — consistency checking of subgroup tables against the overall margin, sharp conditional risk-difference bounds by linear programming, and a five-state verdict taxonomy that keeps inconsistency (`INFEASIBLE`), underspecification (`UNSUPPORTED`), and nonidentification (`REVERSIBLE`) strictly separate; (ii) cost-ordered suggestions of additional *reportable* summaries (re-tabulation of already-collected data before new measurement before new enrollment), each with its assumptions stated and none promising resolution; and (iii) three source-verified applications to landmark trials and a meta-analysis, including a documented real-data direction reversal that unexamined subgroup pooling would have produced. The name IONE also denotes an earlier observational-stratification framework by the same author (here: "original/legacy IONE"); the two share only the name and are distinguished throughout.

The tool does not recommend treatment for any patient, does not claim external validity, and returns abstention (`UNSUPPORTED`) as a first-class output when the information class is not supported. The remainder of the paper defines the audit problem and the exact computation (Section 2), describes the implementation and its verification (Section 3), reports the three real-data audits (Section 4), and discusses scope, limitations, and relation to prior work (Section 5).

2. The audit problem and exact computation

2.1. Data and estimand

We consider the smallest setting in which the problem already bites: a binary stratum indicator $X \in \{0,1\}$ (e.g., "receiving no respiratory support at randomization"), a binary outcome $Y \in \{0,1\}$ (e.g., death within 28 days), and two arms $T \in \{0,1\}$ (control, treatment). The published report supplies arm-level margins $\bar{p}_0 = P(Y{=}1 \mid T{=}0)$ and $\bar{p}_1 = P(Y{=}1 \mid T{=}1)$ — overall event risks per arm — and a subgroup table that supplies, for each named stratum, the arm-specific counts (e_{1x}, n_{1x}) and (e_{0x}, n_{0x}), from which the stratum prevalence $q = P(X{=}1)$ (or arm-specific prevalences q_0, q_1) is estimated. The estimand is the stratum-specific risk difference

$$\mathrm{RD}_x \;=\; P(Y{=}1 \mid T{=}1, X{=}x) - P(Y{=}1 \mid T{=}0, X{=}x), \qquad x \in \{0,1\}.$$

Write the four latent conditional risks as $\theta = (p_{00}, p_{01}, p_{10}, p_{11})$ where $p_{tx} = P(Y{=}1 \mid T{=}t, X{=}x)$. The published margins impose the linear equalities

$$(1-q)\,p_{0,0} + q\,p_{0,1} = \bar{p}_0, \qquad (1-q)\,p_{1,0} + q\,p_{1,1} = \bar{p}_1, \tag{1}$$

and $\theta \in [0,1]^4$. If the report additionally publishes a per-stratum outcome margin $P(Y{=}1 \mid X{=}x)$, the constraint $(1-m_x)\,p_{0x} + m_x\,p_{1x} = P(Y{=}1 \mid X{=}x)$ applies, where $m_x = P(T{=}1 \mid X{=}x)$ must be supplied by the analyst — it is never defaulted to the allocation ratio. Because all constraints are linear and the feasible set Θ is a polytope, the sharp bounds of the linear estimand $\mathrm{RD}_x = p_{1x} - p_{0x}$ are obtained exactly by two linear programs,

$$\mathrm{RD}_x^- = \min_{\theta \in \Theta}(p_{1x} - p_{0x}), \qquad \mathrm{RD}_x^+ = \max_{\theta \in \Theta}(p_{1x} - p_{0x}),$$

solved with the HiGHS dual simplex solver (Huangfu and Hall 2018; Virtanen, Gommers, Oliphant *et al.* 2020). "Sharp" is used in the standard sense: the interval is the tightest one implied by (1) alone; both endpoints are attained by feasible conditional distributions.

Three conditions govern interpretation. First, the bounds are *identification bounds on the published margins*, not confidence intervals: sampling uncertainty in the reported counts is a separate layer that we do not model here (cf. Imbens and Manski 2004). Second, when arm compositions differ ($q_1 \neq q_0$), each arm uses its own stratum prevalence; a single common q is used only when the arm shares are compatible. Third, nothing in (1) restricts unmeasured-effect-modifier structures beyond X; declaring a suspected unmeasured modifier changes only the augmentation advice (Section 2.4), not the bounds.

2.2. Verdicts

Given a decision threshold $\tau \geq 0$ on the adverse-event scale (benefit requires $\mathrm{RD}_x < -\tau$), the audit returns exactly one of five disjoint states:

- `ROBUST_TREAT`: every model in Θ gives $\mathrm{RD}_x < -\tau$;
- `ROBUST_NO_TREAT`: every model in Θ gives $\mathrm{RD}_x > -\tau$;
- `REVERSIBLE`: Θ contains models on both sides of $-\tau$ (genuine nonidentification of the decision);
- `INFEASIBLE`: Θ is empty — the published aggregates are mutually inconsistent;
- `UNSUPPORTED`: the input lacks a required specification (stratum prevalence, arm mixing proportion) or the solver fails.

The separation matters in practice: `INFEASIBLE` points to a reporting error worth querying, `UNSUPPORTED` to missing analyst input, and `REVERSIBLE` to an information gap in the literature — three different actions.

2.3. Consistency checking

Before the LP is run, subgroup counts are checked against the overall table: for exhaustive strata, sums of events and denominators must equal the overall counts; small gaps are flagged as warnings (rounding or an unpublished stratum), while exceedances are errors that render the audit `INFEASIBLE`. Duplicate stratum names are treated as overlap errors. This check is cheap and frequently pays: subgroup tables that cannot support any conclusion are identified before any bounds are computed.

2.4. Reportable augmentation

When the verdict is `REVERSIBLE` or `UNSUPPORTED`, the package lists additional aggregate summaries the original investigators could in principle supply, ordered by expected burden:

1. *re-tabulation* (near-zero cost): an arm × stratum outcome cross-tabulation, or per-stratum baseline risk;
2. *new measurement* on existing cohorts;
3. *new enrollment*: a stratum-restricted randomized cohort;
4. *abstain*: no publishable summary is shown to resolve the decision.

For the binary-stratum model, an arm × stratum cross-tabulation adds two independent equations and collapses Θ to a point — identifiability is structural. Every option is classified (`sufficient_assumed`, `potentially_informative`, `irrelevant`, `infeasible`, `unknown`) together with the assumptions under which its classification holds; no option carries an unconditional guarantee of resolution, since the future cells are unknown a priori.

2.5. Retrospective borrow check

As a demonstration diagnostic — not a borrowing method — the package pools the other subgroups' log risk ratios (fixed effect) and compares the prediction with the held-out subgroup's actual crude risk ratio. The comparison is retrospective by construction: it documents what unexamined pooling would have concluded on already-published data, not prospective validation, and it uses crude rather than adjusted effects.

3. Implementation and verification

The package `ione` (distribution name `ione-audit`, version 0.2.0) is pure Python depending only on NumPy (Harris, Millman, van der Walt *et al.* 2020) and SciPy (Virtanen *et al.* 2020), with the LP solved by HiGHS (Huangfu and Hall 2018). It provides a Python API (`audit_decision`, `suggest_augmentation`, `borrow_check`, `EffectSummary.check_consistency`) and a command-line interface (`ione audit`, `ione borrow-check`) emitting structured JSON. Source, tests, and documentation are at https://github.com/bougtoir/ione-stratification-framework.

Verification is algebraic rather than self-referential. For the two-by-two-by-two model the sharp bounds have a closed form: solving (1) for $p_{t,1}$ given free $p_{t,0}$ and intersecting the resulting windows yields $\mathrm{RD}_1^- = c - k\,d_{\max}$ and $\mathrm{RD}_1^+ = c - k\,d_{\min}$ with $c = (\bar{p}_1 - \bar{p}_0)/q$, $k = (1-q)/q$, and $d_{\min}, d_{\max}$ the range of $p_{1,0} - p_{0,0}$ over the feasible windows. The test suite checks LP optima against this closed form (agreement to 10^{-8} on curated cases; 1.8×10^{-15} worst case over 200 random margins), against coarse full enumeration (sign agreement), and against published effect estimates (crude risk ratios for all three datasets, Haldane-corrected odds ratios for zero cells). Edge cases are asserted explicitly: missing prevalence returns `UNSUPPORTED`, inconsistent margins return `INFEASIBLE`, prevalence near 1 collapses bounds

Table 1: Source-verified published aggregate datasets used for the audits (crude arm-level counts): RECOVERY (RECOVERY Collaborative Group 2021), ISIS-2 (ISIS-2 Collaborative Group 1988), and WHO REACT (WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group and Sterne 2020).

study	outcome	stratum	events/n, treat vs. control
RECOVERY (NEJM 2021)	28-day mortality	overall	482/2104 vs. 1110/4321
		invasive ventilation	95/324 vs. 283/683
		oxygen only	298/1279 vs. 682/2604
		no oxygen	89/501 vs. 145/1034
ISIS-2 (Lancet 1988)	5-week vascular mortality	aspirin overall	804/8587 vs. 1016/8600
		streptokinase arm	791/8592 vs. 1029/8595
		both agents	343/4292 vs. 568/4300
REACT (JAMA 2020)	28-day mortality	overall	222/678 vs. 425/1025
		serious adv. events	64/354 vs. 80/342

Table 2: Decision audits on published margins (threshold $\tau = 0$). "Prevalence" is the stratum share of the respective arm population; arm-specific prevalences are used where arm shares diverge.

study: target stratum	prevalence	bounds on RD	verdict
RECOVERY: invasive ventilation	0.157	[−1.000, +1.000]	REVERSIBLE
RECOVERY: oxygen only	0.604	[−0.425, +0.379]	REVERSIBLE
RECOVERY: no oxygen	0.239	[−1.000, +0.959]	REVERSIBLE
ISIS-2: streptokinase arm	1.000	[−0.025, −0.025]	ROBUST_TREAT
ISIS-2: both agents	0.500	[−0.236, +0.187]	REVERSIBLE
REACT: serious adv. events	0.334/0.522[a]	[−1.000, +0.627]	REVERSIBLE

[a] arm-specific prevalences q_0, q_1.

to the marginal risk difference, and prevalence near 0 widens them to $[-1, 1]$. All 27 tests pass under Python 3.9–3.12 in continuous integration.

4. Applications to published trials

Table 1 summarizes the three built-in datasets; all counts are transcribed from the primary publications and cross-verified (the RECOVERY counts match the primary report (RECOVERY Collaborative Group 2021) percentage-for-percentage; ISIS-2 from ISIS-2 Collaborative Group (1988); REACT from WHO Rapid Evidence Appraisal for COVID-19 Therapies (REACT) Working Group and Sterne (2020)). We report crude estimates only; where the original study's headline was model-adjusted (REACT's fixed-effect odds ratio), that is noted rather than reproduced.

Table 2 reports the decision audits. The dominant finding is `REVERSIBLE`: for every non-degenerate subgroup target in the three datasets, the published margins admit compatible models on both sides of the decision threshold, and for minority strata the identified interval spans nearly the full $[-1, 1]$ range (Figure 1). ISIS-2 yields the single `ROBUST_TREAT` verdict, but it is degenerate: the streptokinase "subgroup" coincides with the whole arm population ($q = 1$), so the conditional risk difference equals the marginal one — the audit correctly reports robustness only where the margins actually pin it down. No published dataset was `INFEASIBLE` or `UNSUPPORTED`; Figure 2 shows those states on labelled constructed inputs.

The borrow check illustrates why the audit matters. On RECOVERY, pooling the other subgroups (invasive ventilation and oxygen only) predicts benefit for the untreated subgroup —

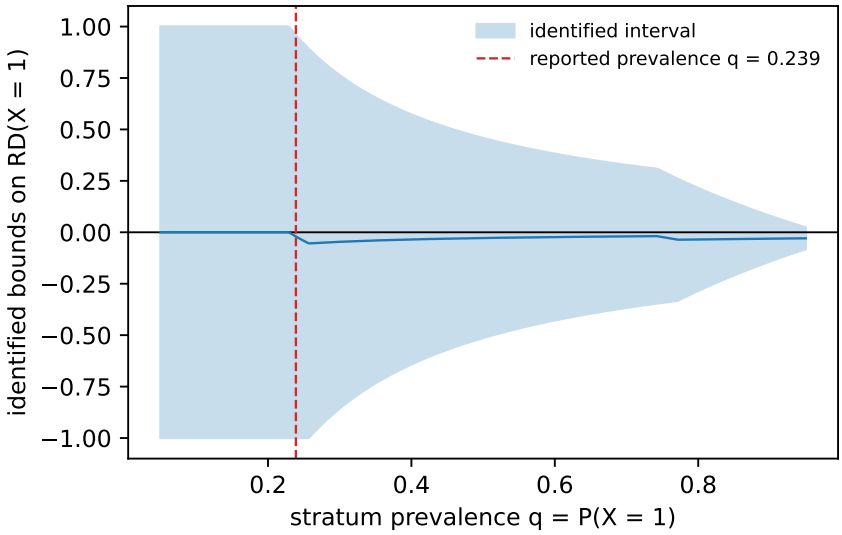


Figure 1: Identified bounds on the conditional risk difference for the RECOVERY overall margins as a function of the stratum prevalence q. The dashed line marks the reported prevalence of the "no oxygen" subgroup ($q = 0.239$). Bounds widen toward $[-1, 1]$ for rare strata — the information loss is driven by prevalence, not by sample size as modeled here.

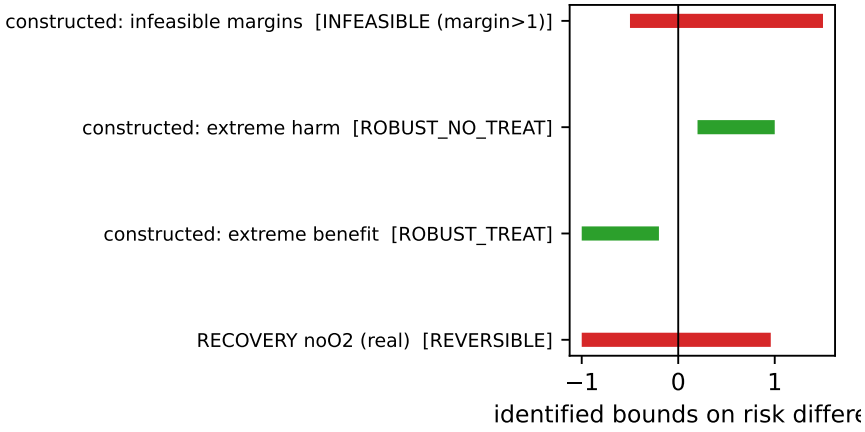


Figure 2: Identified bound intervals across verdict states. The RECOVERY "no oxygen" audit is real data; the robust and infeasible rows are labelled constructed inputs (extreme margins; margin > 1). `UNSUPPORTED` (missing specification) produces no bounds and is not shown.

risk ratio 0.835 (95% CI 0.755–0.923) — while the actual published no-oxygen estimate shows harm, risk ratio 1.267 (0.995–1.613), with non-overlapping intervals. This is a retrospective demonstration on real published counts, not prospective validation of any borrowing rule; it documents that direction reversal under unexamined pooling is not hypothetical.

For the `REVERSIBLE` RECOVERY cases, the augmentation list begins with a zero-cost option: an arm × respiratory-support outcome cross-tabulation of already-collected trial data, which would collapse the identified interval to a point if produced (the future cell values remain unknown, so the post-information decision is not guaranteed — the classification states this). Per-stratum baseline risk and a stratum-restricted randomized cohort follow in burden order.

Audits run at interactive speed: each LP pair solves in under a millisecond, and the full reproduction script (200 random-margin checks, a 400-point prevalence sweep, all audits and figures) completes in seconds on a laptop.

5. Discussion

5.1. What the audit adds

The contribution of `ione` is a reproducible statistical audit workflow, not a new estimator: partial identification of conditional quantities from aggregate margins is classical (Cross and Manski 2002; Manski 2003; King 1997), the LP formulation follows Balke and Pearl (1997), and the same marginal-consistency program has recently been developed independently for economic outcomes (Moon 2026). What was missing, and what the package supplies, is an applied layer that a clinical researcher can run on a published report: consistency screening of the printed tables, sharp bounds with a decision-verdict taxonomy that separates "the data contradict themselves" from "the data do not decide" from "you have not given me enough specification," and an ordered list of what could still be reported to narrow the answer — starting with re-tabulations that cost the original investigators nothing.

5.2. Practical uses and the value of abstention

Three uses appear immediate. First, citation hygiene: a clinician citing a subgroup result can check whether the published margins admit the implied decision at all; the RECOVERY no-oxygen case shows that widely interpreted subgroup tables can be fully decision-reversible. Second, requests for re-analysis: the augmentation ordering converts "we need IPD" — usually a dead end — into a concrete, minimal, zero-cost request (an arm × stratum cross-tabulation) that trial groups can often supply. Third, a disciplined abstention state: `UNSUPPORTED` is an output, not an error, so incomplete reports end in an explicit statement of what is missing rather than silence.

5.3. Limitations

The current release handles binary strata, binary outcomes, and two arms only; richer covariates enlarge the polytope and are future work. Bounds are identification bounds on the published margins treated as exact — sampling uncertainty in the reported counts is not propagated, and valid confidence sets for partially identified parameters are a separate known layer (Imbens and Manski 2004). Estimands are crude: adjusted published estimates cannot be reproduced from aggregate counts and are never substituted for them. The stratum prevalence (or arm-specific prevalences) must be estimable from the report, which restricts applicability to trials that publish subgroup denominators — that availability is itself the binding real-world constraint. The borrow check is a retrospective diagnostic, not a validated borrowing method, and crude risk ratios may differ from adjusted ones. Finally, the audit audits identifiability, not truth: biased underlying estimates produce faithful bounds around biased margins.

5.4. Relation to further work

The framework complements, rather than competes with, sensitivity analysis (VanderWeele and Ding 2017) and transportability (Pearl and Bareinboim 2014): those approaches ask whether conclusions survive bias or transfer across populations under assumptions, while the audit asks what the published numbers alone identify under the stated model. Extensions worth pursuing are interval-valued margins that carry sampling error, multi-level and continuous strata, and joint audits across linked trial reports.

6. Conclusions

Published aggregate trial data carry more decision-relevant information than they are usually credited with — and less than subgroup headlines suggest. `ione` makes the gap explicit and

reproducible: sharp identification bounds on subgroup decisions, a verdict taxonomy that never conflates inconsistency, underspecification, and nonidentification, and a cost-ordered path to what could still be reported. On three landmark datasets the published subgroup margins are mostly decision-reversible, and a retrospective borrow check reproduces a real direction reversal that unexamined pooling would have produced. The practical value is a disciplined "we do not know, and here is the cheapest thing that would tell us" instead of either over-claiming or silence.

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